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DIFFERENCING ASYMPTOTIC DIFFUSION THEORY

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Differencing Asymptotic Diffusion Theory*

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ABSTRACT

A diffusion theory is presented which extends asymptotic diffusion to non-uniform material properties. Finite difference methods for the diffusion theory naturally result in jump conditions on interfaces when appropriate.

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Standard diffusion theory can fail in several ways. If the mean free path is long compared to scale heights or geometrical distances then it is inappropriate and leads to several kinds of errors. Some of these, such as photons moving faster then c, can be "fixed" with flux limiters or more appropriately calculated by using Sn, Pn or MC transport. Even if the mean free path is short, however, standard diffusion theory will fail if the net absorption or creation rate is not small compared to the scattering rate. Asymptotic diffusion theory addresses this situation.

Actually, asymptotic diffusion is only valid in a uniform medium. I assume that each zone is a uniform medium and match forward and backward fluxes at the interfaces between zones. The formulas thus produced have several interesting properties. In the limit of small zones the method does NOT reduce to asymptotic diffusion. Rather it results in a diffusion theory that is very similar to the "new diffusion" derived by Pomraning (1973). Several authors have pointed out the importance of jump boundary conditions at interfaces where source rates or absorption properties change abruptly. The method derived here produces these jump conditions as a natural consequence when they are appropriate.

I begin by presenting, without proof, the principle results of asymptotic diffusion. The reader is referred to Pomraning (1973) and Winslow (1968) for a more complete discussion. Only a single group formulism is considered, but the generalization to multi-group is straightforward.

In slab geometry the steady state particle distribution in the asymptotic region (far from boundaries) is given by

$$\phi_0 = Ae^{\alpha\sigma x} + Be^{-\alpha\sigma x} \tag{1}$$

where $\sigma = 1/\lambda$ is the total cross-section, $\phi_0 = nv$ is the isotropic particle flux and α is given by

$$\alpha = \tanh\left(\frac{\alpha}{\omega}\right) \tag{2}$$

Here ω is the effective scattering albedo including sources

$$\omega = \frac{1}{\sigma}\left[\sigma - \sigma_{abs} + \frac{S}{\phi_0}\right] \tag{3}$$

For photonics problems σ has already been corrected for stimulated emission, while for neutronics σ includes the transport correction and S includes the extra neutrons produced by fission, n2n, etc. For neutronics problems in which the neutron density increases exponentially with time we follow Winslow (1968) and add α_N/v to σ, where α_N is the problem averaged exponential growth rate.

Cleary $\omega = 1$ for pure scattering, $\omega = 0$ for pure absorption and $\omega \to \infty$ for strong sources. The net particle flux in the asymptotic region is

$$\phi_1 = -\frac{D_0}{\sigma} \nabla \phi_0 \tag{4}$$

where

$$D_0 = \frac{1-\omega}{\alpha^2} \tag{5}$$

is the dimensionless diffusion coefficient of ($D_0 = 1$ for $\omega = 0$, $D_0 = 1/3$ for $\omega = 1$ and $D_0 \to 4/(\Pi^2 \omega)$ as $\omega \to \infty$). The partial fluxes in the $\pm$ x directions are

$$\phi \pm = \tfrac{1}{4}\mu_0\phi_0 \pm \tfrac{1}{2}\phi_1 \tag{6}$$

where

$$\mu_0 = \frac{\omega}{2\alpha^2} \ln\left(\frac{1}{1-\alpha^2}\right) \tag{7}$$

($\mu_0 = 1$ for $\omega = 0$, $\mu_0 = \tfrac{1}{2}$ for $\omega = 1$ and $\mu_0 \to 4/(\Pi^2\omega) \ln(\frac{\Pi}{2}\omega)$ as $\omega \to \infty$), when $\omega > 1$ it is convenient to introduce $\beta = -i\alpha$, in which case Equations (2), (5) and (7) become

$$\beta = \tan\left(\frac{\beta}{\omega}\right) \tag{2'}$$

$$D_0 = \frac{\omega - 1}{\beta^2} \tag{5'}$$

$$\mu_0 = \frac{\omega}{2\beta^2} \ln(1 + \beta^2) \tag{7'}$$

These are the principle results of asymptotic diffusion theory. The functions described by Equations (2), (5) and (7) can be approximated to 5% accuracy by

$$\alpha^2 = (1 - \omega)\ (1 + \omega + \omega^2)\ (\text{for } \omega < 1) \tag{2"}$$

$$D_o = \frac{.4674 + \omega}{.4674 + 1.4674\,\omega + 2.4674\,\omega^2} \tag{5"}$$

$$\mu_o = \frac{\ln\,(8.3548 + 1.5708\,\omega)}{2.1228 + 2.4674\,\omega} \tag{7"}$$

Next we postulate an interface located at X = 0 with zone A to the left, and B to the right. Within each zone the total cross-section and scattering albedo are assumed constant. Applying Equation (6) at the interface and forming the quantities $\phi_+ + \phi_-$ and $\phi_+ - \phi_-$ yields

$$\mu_o{}^A\ \phi_o{}^A(0) = \mu_o{}^B\ \phi_o{}^B(0) \tag{8}$$

$$\phi_1{}^A(0) = \phi_1{}^B(0) \tag{9}$$

Equation (8) shows that $\mu_o\phi_o$ is continuous across the interface while Equation (9) shows particle conservation. Applying Equations (8), (9) on all zonal interfaces and Equation (4) within each zone constitutes our method. It reduces to asymptotic diffusion when μ_o and D_o are constants and to normal diffusion when $\omega = 1$.

In order for this method to be useful, we must write the interface flux, ϕ_1 (0), in terms of the zone centered particle densities $\phi_o^A(-\Delta X_A)$ and $\phi_o^B(\Delta X_B)$. Here $\Delta X_{A,B}$ are the zone half widths. For very small zones ($\alpha\sigma\Delta X << 1$) we can rewrite Equation (1) as

$$\phi_o = A'(e^{\alpha\sigma X} + e^{-\alpha\sigma X}) + B'(e^{\alpha\sigma X} - e^{-\alpha\sigma X}) \tag{10}$$

$$\simeq 2A' + 2\alpha\sigma X B'$$

showing that ϕ_o is linear in X. Equation (4) then yields

$$\phi_o^A(0) = \phi_o^A(-\Delta X_A) - \frac{\sigma_A \Delta X_A}{D_o^A} \phi_1(0) \tag{11a}$$

$$\phi_o^B(0) = \phi_o^B(\Delta X_B) + \frac{\sigma_B \Delta X_B}{D_o^B} \phi_1(0) \tag{11b}$$

Using these expressions in Equation (8) and solving for $\phi_1(0)$ yields

$$\phi_1(0) = \frac{\mu_o^A \phi_o^A(-\Delta X_A) - \mu_o^B \phi_o^B(\Delta X_B)}{\frac{\mu_o^A}{D_o^A} \sigma_A \Delta X_A + \frac{\mu_o^B}{D_o^B} \sigma_B \Delta X_B} \tag{12}$$

which is equivalent to the diffusion equation

$$\phi_1 = -\frac{D_o}{\mu_o} \nabla(\mu_o \phi_o) \tag{13}$$

By comparison, Pomraning (1973) derived a diffusion theory

$$\phi_1 = -\frac{1}{\sigma} \nabla(D_0\phi_0) \tag{14}$$

which has properties very similar to Equation (13).

Equation (12) is only valid for small zones ($\alpha\sigma\Delta X \ll 1$). In general ϕ_0 will not be linear in X, but is given by Equation (1). In order to introduce this exponential dependence within the framework of a diffusion model we modify the optical depth of a zone half-width by a function $\chi(\alpha\sigma\Delta X)$. Equation (11b) then becomes

$$\phi_0(0) = \phi_0(\Delta X) + \frac{\sigma\Delta X}{D_0} \chi(\alpha\sigma\Delta X)\, \phi_1(0) \tag{11b'}$$

where we have dropped the zonal subscript, B. Clearly $\chi(y)$ must approach unity as $y \to 0$ since Equation (11b) is valid for small zones. The form of χ is determined by forcing Equation (11b') to yield the correct steady-state flux, $\phi_1(0)$, in the extreme case where the exact solution is a pure exponential (A = 0 in Equation (1)). Under these conditions application of Equation (4) at X = 0 yields the exact asymptotic diffusion result,

$$\phi_1(0) = \alpha D_0\, \phi_0(0) \tag{15}$$

By comparison, the finite difference form of the steady-state diffusion equation is

$$0 = \frac{d\phi_0(\Delta X)}{dt} = \frac{1}{2\Delta X}(\phi_1(0) - \phi_1(2\Delta X)) - (1-\omega)\sigma\phi_0(\Delta X) \tag{16}$$

Applying Equation (11b') at $X = 0$ and its equivalent at $X = 2\,\Delta X$ results in

$$\phi_1(0) = \frac{D_0}{\sigma\Delta X\ \chi(\alpha\sigma\Delta X)}\left(\phi_0(0) - \phi_0(\Delta X)\right) \tag{17a}$$

and

$$\phi_1(2\Delta X) = \frac{D_0}{\sigma\Delta X\ \chi(\alpha\sigma\Delta X)}\left(\phi_0(\Delta X) - \phi_0(2\Delta X)\right) \tag{17b}$$

where

$$\phi_0(2\Delta X) = \phi_0(0)\ e^{-2\alpha\sigma\Delta X}$$

Together with Equations (5) and (16) this gives

$$\phi_0(\Delta X) = \frac{\tfrac{1}{2}\phi_0(0)\ (1 + e^{-2\alpha\sigma\Delta X})}{1 + (\alpha\sigma\Delta X)^2\ \chi(\alpha\sigma\Delta X)} \tag{18}$$

and thus from Equation (17a)

$$\phi_1(0) = \alpha D_0\phi_0(0)\ \frac{\tfrac{1}{2}(1-e^{-2\alpha\sigma\Delta X}) + (\alpha\sigma\Delta X)^2\chi(\alpha\sigma\Delta X)}{[\alpha\sigma\Delta X\ \chi(\alpha\sigma\Delta X)]\ [1+(\alpha\sigma\Delta X)^2\chi(\alpha\sigma\Delta X)]} \tag{19}$$

Equations (15) and (19) determine the function χ to be given by

$$1 = \frac{1}{y\chi(y)}\ \frac{\tfrac{1}{2}(1-e^{-2y}) + y^2\chi(y)}{1 + y^2\chi(y)} \tag{20}$$

from which it can be shown that $\chi(y) \to 1 - 1/3\ y^2$ as $y \to 0$ and $\chi(y) \to 1/(y+\tfrac{1}{2})$ as $y \to \infty$. Equation (20) can be fit to 7% by

$$\chi(y) = \frac{1 + y}{1 + y + y^2} \tag{20'}$$

Using Equation (11b') and its equivalent for zone A in Equation (8) yields the flux between zones A and B

$$\phi_1(0) = \frac{\mu_0^A \phi_0^A(-\Delta X_A) - \mu_0^B \phi_0^B(\Delta X_B)}{\frac{\mu_0^A}{D_0^A}\, \sigma_A \Delta X_A\, \chi(\alpha_A \sigma_A \Delta X_A) + \frac{\mu_0^B}{D_0^B}\, \sigma_B \Delta X_B\, \chi(\alpha_B \sigma_B \Delta X_B)} \tag{12'}$$

where we use $\alpha = 0$ for $\omega > 1$. Since $\chi \to 1$ as $\alpha\sigma\Delta X \to 0$ we see that Equation (12') reduces to Equation (12) for small zones.

An interesting application of Equation (12') is the case where zone A is thin and zone B is a thick perfect absorber ($\mu_0 = D_0 = \alpha = 1$). Since for $y \to \infty$, $\chi(y) \to 1/y$, we obtain

$$\phi_1(0) = \frac{\mu_0^A \phi_0^A(-\Delta X_A)}{\frac{\mu_0^A}{D_0^A}\, \sigma_A \Delta X_A + 1} \tag{21}$$

or

$$\phi_1(0) = \frac{D_0^A \phi_0^A(-\Delta X_A)}{\sigma_A \Delta X_A + \frac{D_0^A}{\mu_0^A}} \tag{22}$$

The quantity D_o/μ_o is 2/3 when zone A is a pure scatterer and 1.0 when a pure absorber. Thus Equation (12') reduces to the well known Milne boundary (jump) conditions at the surface of a perfect absorber. This equivalence between vacuum and a perfect absorber is obvious on physical grounds. The f ct that it is produced as a natural consequence of Equation (12') is an important difference between this and other diffusion methods. Normally these conditions must be imposed arbitrarily on certain interfaces. This is probably acceptable for neutronics problems, but for photonics the absorption and scattering properties of materials are time dependent, making it impossible to impose the appropriate jump conditions. There has been some success in using flux-limiters to establish the jump conditions, but this technique fails unless the absorption region is finely zoned ($\alpha\sigma\Delta X << 1$).

In order to test this diffusion method under more realistic conditions I have run several one dimensional, plane parallel, 10-group neutronics problems. In these problems a 2cm thick slab of U^{235} at density 50 g/cc was sandwiched between two 2 cm thick slabs of Be^7 at density 200 g/cc. The Be^7 is thick enough so that no neutrons are transmitted while its scattering albedo as a strongly increasing function of neutron energy. The primary diagnostic from these calculations is the neutron exponentiation rate, α_N, expressed in sh^{-1}. Calculations were run with four different transport methods: sixteen angle Sn (S16), standard diffusion (SD), asymptotic diffusion (AD) and the finite difference of asymptotic diffusion presented in this paper (FDAD). Each method was applied to four problems: (1) no Be^7 "bread" on the sandwich, (2) Be^7 zoned with 4 zones of 0.5 cm each, (3) Be^7 zoned with 10 zones of .05 cm next to the U^{235} followed by 3 zones of 0.5 cm, (4) Be^7 zoned with 10 zones of .01 cm next to the U^{235} followed by 8 zones of .05 cm followed by 6 zones of .25 cm. Table I shows the results of these calculations.

Although the S16 calculations of the finely zoned Be^7 did not settle, they at least indicate that $\alpha_N \approx 0$. The coarsely zoned S16 calculation is expected to produce too small an α_N since it cannot account adequately for neutron reflection from the Be^7. The FDAD method detailed in this paper also produces $\alpha_N \approx 0$, reasonably independent of zoning. The zoning dependence is probably due to the multi-group formalism which invalidates our assumption of uniform ω within a zone.

Both standard and asymptotic diffusion produce erroneous results for all zoning patterns of the Be^7 although they do reasonably well for the bare U^{235}. As the optical thickness of the first Be^7 zone becomes large both SD and AD will treat the Be^7 as a perfect reflector and thus α_N will approach its infinite medium value of 3.265.

SUMMARY

In summary, a diffusion theory has been presented which extends asymptotic diffusion to non-uniform material properties. The method is similar to Pomraning's "new diffusion". Finite difference methods for the diffusion theory naturally result in jump conditions on interfaces when they are appropriate.

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Table I

Neutron Exponentiation Rate for U^{235} - Be^7 Sandwich

Problem #

	1	2	3	4
S16	-.64	-.53	-.113*	.0078*
SD	-.81	2.67	.36	-.47
AD	-.68	2.96	1.99	1.74
FDAD	-.61	.13	.078	-.044

*These calculations did not settle in 500 cycles.